

Project 01 - Online Food Ordering System

Overview

A multi-sided food delivery platform connecting customers, restaurant managers, and delivery agents under a single platform admin. Customers browse restaurants and menus, place orders, and track deliveries in real time. Restaurant managers run their full operations through the platform, and delivery agents manage their own assignment queue and earnings. The platform admin oversees all parties and maintains platform health.

Roles

Role	Responsibility
Customer	Browse, order, track, review
Restaurant Manager	Menu, orders, reviews, analytics
Delivery Agent	Accept deliveries, update status, track earnings
Platform Admin	User management, platform oversight, reports

Shared Database Schema

All four students create and use the same database. The schema must include at minimum the following tables:

users — All platform users (customers, managers, agents, admins) identified by role

- id, name, email, password_hash, phone, role, profile_pic, is_active, created_at

restaurants — Restaurant profiles managed by restaurant managers

- id, manager_id, name, description, cuisine_type, address, city, logo_path, opening_hours, delivery_radius_km, is_open, is_approved, created_at

menu_categories — Categories within a restaurant's menu

- `id, restaurant_id, name, display_order`

menu_items — Individual items on a restaurant's menu

- `id, restaurant_id, category_id, name, description, price, image_path, is_available, created_at`

discounts — Special offers created by restaurant managers on specific items

- `id, menu_item_id, restaurant_id, discount_pct, valid_from, valid_until, is_active`

orders — Customer orders placed through the platform

- `id, customer_id, restaurant_id, agent_id, delivery_address, payment_method, subtotal, delivery_fee, total_amount, status (pending/accepted/preparing/ready/picked_up/delivered/cancelled), estimated_delivery_minutes, created_at`

order_items — Line items within each order

- `id, order_id, menu_item_id, quantity, unit_price`

delivery_agents — Extended profile for delivery agents

- `id, user_id, vehicle_type, is_online, current_location_text, total_earnings, is_approved`

delivery_assignments — Links an agent to an order delivery

- `id, order_id, agent_id, assigned_at, picked_up_at, delivered_at, status`

reviews — Customer reviews for restaurants

- `id, order_id, customer_id, restaurant_id, rating (1–5), comment, manager_reply, created_at`

saved_restaurants — Customer wishlist of favourite restaurants

- `id, customer_id, restaurant_id, created_at`

delivery_addresses — Saved delivery addresses per customer

- `id, customer_id, label, address_line, city, is_default`

complaints — Customer or agent complaints submitted to admin

- `id, submitter_id, subject, description, status (open/resolved), created_at`

platform_settings — Key-value store for admin-controlled platform configuration

- `id, setting_key, setting_value`

Each group member is responsible for creating their own data in the database. Each member may only insert the data that is required for demonstrating their own work

Technical Requirements

- Follow the **MVC pattern**: separate Models, Views, and Controllers; no business logic in view files
 - Use **PHP** for all server-side logic and **MySQL** as the database
 - All database queries must use **mysqli with prepared statements** — no raw string insertion allowed
 - Authentication must use **PHP sessions** with role-based access control on every protected page
 - Each student must implement **at least one feature using AJAX** (XMLHttpRequest) communicating with a PHP API endpoint that returns JSON
 - Maintain a central repository for the group where each student will push their code. It is the responsibility of each group member to ensure their code works when cloned and run in a XAMPP Apache server.
 - Use **Git** on the command line for version control. Do not push to main/master branch directly. Create feature branches and submit a Pull Request for each major feature
 - Submit: (1) working online codebase, (2) hardcopy report describing your role's features
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Role 1 — Customer

Description

The Customer is the primary end-user of the platform. They discover restaurants, build and place orders, monitor deliveries, and engage with the platform through reviews and saved preferences. The Customer experience spans discovery, transaction, post-order engagement, and account management.

Features

- Register with name, email, phone, and password; log in and log out securely
- Manage profile: update personal information, change password, upload profile picture
- Manage multiple saved delivery addresses (add, edit, delete, set default)
- Browse all approved restaurants; search by name, cuisine type, or city; filter by rating or cuisine
- View restaurant detail page: logo, description, cuisine, opening hours, average rating, and full menu grouped by category
- Browse menu items within a restaurant; view item details including price, description, and image
- Add items to a session-based cart; update item quantities; remove items; view running total
- Apply active discount offers automatically when adding discounted items to cart
- Proceed to checkout: confirm delivery address (select from saved or enter new), choose payment method (Cash / Card), and review order summary before placing
- View order confirmation page with order ID and estimated delivery time
- Track active order status in real time — status badge updates automatically via AJAX polling without a full page reload
- View complete order history; re-order a previous order (adds all its items to cart with one click)

- Cancel a pending order before the restaurant has accepted it
 - Submit a star rating (1–5) and written review for a restaurant after an order is delivered; view own past reviews
 - Save and unsave restaurants to a personal favourites list; browse the favourites list
 - Submit a complaint to the platform admin regarding an order or delivery experience
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Role 2 — Restaurant Manager

Description

The Restaurant Manager operates a single restaurant through the platform. They configure their restaurant profile, manage the full menu, handle incoming orders through their lifecycle, respond to customer feedback, and monitor their business performance through built-in analytics.

Features

- Register a restaurant account; submit restaurant details for admin approval; log in after approval
 - Manage restaurant profile: name, description, cuisine type, address, city, logo upload, opening hours, delivery radius, and open/closed toggle
 - Manage menu categories: create, rename, reorder, and delete categories
 - Manage menu items: add items (name, description, price, image, category, availability toggle), edit existing items, delete items
 - Create limited-time discount offers on specific menu items: set discount percentage and validity dates; activate or deactivate discounts
 - View incoming orders in real time via AJAX — new orders appear without a page refresh
 - Accept or reject incoming orders; update order status through the lifecycle (Preparing → Ready for Pickup)
 - View active orders dashboard showing all orders currently in progress grouped by status
 - View full order history with customer name, items ordered, total amount, and delivery status
 - View and respond to customer reviews (post a public reply to any review for their restaurant)
 - View sales analytics: total orders per day/week/month, total revenue, most ordered items, average order value
 - View a list of discount campaigns and their performance (how many orders used the discount)
 - View all complaints submitted to admin related to their restaurant
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Role 3 — Delivery Agent

Description

The Delivery Agent is an independent contractor on the platform. They manage their own availability, pick up delivery assignments for ready orders near them, update delivery progress, and track their earnings and history.

Features

- Register as a delivery agent (name, phone, vehicle type); submit for admin approval; log in after approval
- Manage profile: update personal info, vehicle type, and profile picture; toggle online/offline availability
- View available delivery assignments — orders that are Ready for Pickup and unassigned in the system
- Accept a delivery assignment; view full delivery details: restaurant name and address, customer delivery address, order items list, and any special instructions
- Decline an assignment (returns it to the available pool)
- Update delivery status in sequence: Picked Up → On the Way → Delivered; each update reflected immediately via AJAX on the customer's tracking page
- View current active delivery details at all times while a delivery is in progress
- View complete delivery history: past deliveries with date, restaurant, customer area, and earning per delivery
- View earnings summary: total earned today, this week, this month, and all-time
- View performance stats: total deliveries completed, average delivery time, any complaints logged against them
- Receive a visual notification (AJAX-driven) when a new assignment becomes available while they are online

Role 4 — Platform Admin

Description

The Platform Admin is the super-user who governs the entire platform. They approve new restaurants and agents, manage all accounts, monitor platform-wide activity, configure global settings, and handle disputes. The Admin role has read and write access to every entity in the system.

Features

- Log in to a dedicated admin dashboard showing key platform metrics: total active restaurants, total orders today, total registered users, total active delivery agents

- Manage restaurant accounts: view all restaurants (pending approval and approved), approve or reject new registrations, suspend or reactivate existing restaurants
 - Manage customer accounts: search, view, deactivate or reactivate any customer
 - Manage delivery agent accounts: view all agents (pending and approved), approve or reject new agents, deactivate agents
 - Manage platform-level cuisine categories used to classify restaurants
 - View all orders across the platform with filters by status, date range, restaurant, and customer
 - View and resolve customer complaints: read complaint details, mark as resolved, and add an admin note
 - Manage featured restaurants shown on the customer's homepage (add/remove from featured list)
 - Configure platform settings: commission rate per restaurant, delivery fee structure, estimated delivery time formula
 - View platform-wide analytics: total revenue (sum of all orders), orders per status, top-performing restaurants, busiest delivery agents, peak ordering hours
 - View delivery performance report: average delivery time, on-time rate, failed deliveries
 - Generate and view monthly summary reports for the entire platform
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Submission Checklist

- Each group member is responsible that each of their roles are individually accessible with their own login and dashboard
- Role-based access control prevents any user from accessing another role's pages
- The shared database schema is implemented correctly and consistently across all roles
- At least one AJAX-based feature is implemented per role
- All forms include server-side validation with descriptive error messages
- Git history shows feature branches and pull requests for major features
- Hardcopy report describes all features for your assigned role

Separation of Concerns

- Each group member is responsible for their own code and submission. Other members work will not affect one's development and submission.
- DO NOT rely on any other of your group member for anything, including DB table creation, data insertion, session management etc.